Understanding Python Dataframe's Data Types

by Shahriar Analytical Academy

What Are Data Types in a DataFrame?

1

Definition

Data types (also called **dtypes**) represent the kind of data stored in each column of a DataFrame. It determines how data is stored, interpreted, and processed.

2

Why It Matters

Data types affect memory usage and performance.

They determine which operations can be applied to the data.

Proper data types ensure accurate statistical and mathematical analysis.

3

Real-Life Analogy

Think of data types like labels on boxes — one for numbers, one for dates, one for names. If you put a date in a name box, you won't know how to sort it properly.

Why You Should Care About Data Types

1

Data Integrity

Ensures values are interpreted correctly. E.g., "100" as a string vs 100 as a number.

2

Performance

Numerical types like int64 and float64 are faster and use less memory than object.

3

Analytical Accuracy

Correct types are essential for accurate statistical functions like mean, sum, or correlation.

4

Avoiding Errors

Wrong types may break functions. For example, calculating the average of text values leads to errors.

int64 – Integer Data Type

What It Is

int64 stands for 64-bit integers. It stores **whole numbers** without decimals, including both positive and negative values.

Examples

1, 0, -35, 4000, 1234567890

When It's Used

- Counting items (e.g., number of orders)
- Storing IDs (if not mixed with letters)
- Recording age or quantity

Benefits

Efficient in memory and supports mathematical operations like addition, subtraction, etc.

Important Note

If there's even one missing or non-numeric value, Pandas may default to float64 or object.

float64 – Floating Point (Decimal) Data Type

What It Is

float64 stores **decimal numbers**, including very small and very large values, with double precision.

Examples

3.14, 0.0, -9.81, 12345.6789

When It's Used

- Prices and amounts
- Percentages and probabilities
- Scientific measurements

Key Insight

Always prefer float64 when fractional data is involved.
Operations like averaging, rounding, or scaling require floats.

object – Text or Mixed-Type Data

What It Is

The most flexible data type. Typically used for **strings**, but can also hold mixed content (numbers + text).

Examples

"Dhaka", "apple", "123ABC", "Male"

When It's Used

- Names, addresses, cities
- Descriptive text (e.g., product description)
- Alphanumeric codes (e.g., ID#A03)

Important Note

Object columns use more memory and cannot be used in numerical operations without conversion.

bool – Boolean Data Type

What It Is

Represents only two values: True or False.

1

Examples

True, False

2

When It's Used

- Binary flags (e.g., "Is VIP customer?")
- Results of logical comparisons
- Filtering and condition-based selections

3

datetime64 – Date and Time Data Type

What It Is

1

Stores temporal data — **dates, times, or both** — in an efficient format that supports operations like comparison, subtraction, or formatting.

Examples

2

"2025-04-24", "2023-01-01 15:30:00"

When It's Used

3

- Timestamps (e.g., login time, order date)
- Time series analysis
- Date arithmetic (e.g., age, duration)

Summary and Key Takeaways

1

Data Type Definition

Data types define how each column behaves in a DataFrame.

2

Common Types

Common types include int64, float64, object, bool, datetime64, and category.

3

Data Cleaning

Checking and fixing data types is one of the first steps in data cleaning.

4

Benefits

Correct types improve performance, memory efficiency, and reduce errors.

Thank You!

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